Binary Classification Model Using Logistic Regression

Full Source Code

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

data = load_breast_cancer()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 2))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Full Output

```
Accuracy: 0.98
Confusion Matrix:
[[ 61   2]
 [  2 106]]
Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.97      0.97        63
           1       0.98      0.98      0.98       108

    accuracy                           0.98       171
   macro avg       0.97      0.97      0.97       171
weighted avg       0.98      0.98      0.98       171
```

How to run: Install Scikit-learn with `pip install scikit-learn`. Save the code as `binary.py` and press F5 in Python IDLE.

Note: Output shown is from this dataset and train-test split; results can vary slightly across software versions.